

Specification-Driven Two-Stage CMOS OTA

A reproducible 1.8 V Miller-compensated design, from device characterization through 13-point PVT verification and explicit failure reporting.

The final M1-M10 OTA uses a single external 10 uA reference, a 3 pF Miller capacitor, and a 2 kohm series nulling resistor. All six frozen core metrics pass their hard and stretch targets at every required PVT point. The full nominal characterization also exposes two unresolved requirements: PSRR and the 1.3 V high end of ICMR.

65.54 worst A0 (dB)	14.55 worst UGB (MHz)	66.25 worst PM (deg)
302.71 max power (uW)	7.98 worst SR+ (V/us)	11.20 worst SR- (V/us)

Qualification snapshot

Scope	Result	Frozen requirement	Status
13-point core PVT	A0 / UGB / PM / power / SR+ / SR-	all 13 points	PASS
Nominal settling	0.0747 us	<= 1.5 us	PASS
Nominal CMRR @ 1 kHz	71.32 dB	>= 55 dB	PASS
Nominal output swing	0.18-1.63 V	contains 0.3-1.5 V	PASS
Nominal PSRR+ / PSRR-	36.33 / 36.25 dB	>= 45 dB each	FAIL
Nominal ICMR	0.76-1.22 V	contains 0.8-1.3 V	FAIL

Truth boundary

These are transistor-level **schematic simulations**, not fabricated, measured, post-layout, or Monte Carlo results. The external IREF is ideal. Failed requirements remain visible and are not promoted to passes.

Evidence bundle: 171 completed ngspice decks; Day 4 manifest 6b90c7326a881914 . . .

1. Architecture and sizing

An NMOS differential pair and PMOS mirror load form the first stage. M6/M7 form the common-source output stage. M8-M10 translate one external reference current into the PMOS and NMOS bias voltages. The compensation branch is strictly $V_X \rightarrow R_Z \rightarrow CC \rightarrow V_{OUT}$; loop-break and load elements live only in testbenches.

SKY130A 1.8-V Two-Stage Miller-Compensated CMOS OTA

Frozen Day 3 design • single 10- μ A reference • schematic-level simulation • device dimensions are W/L

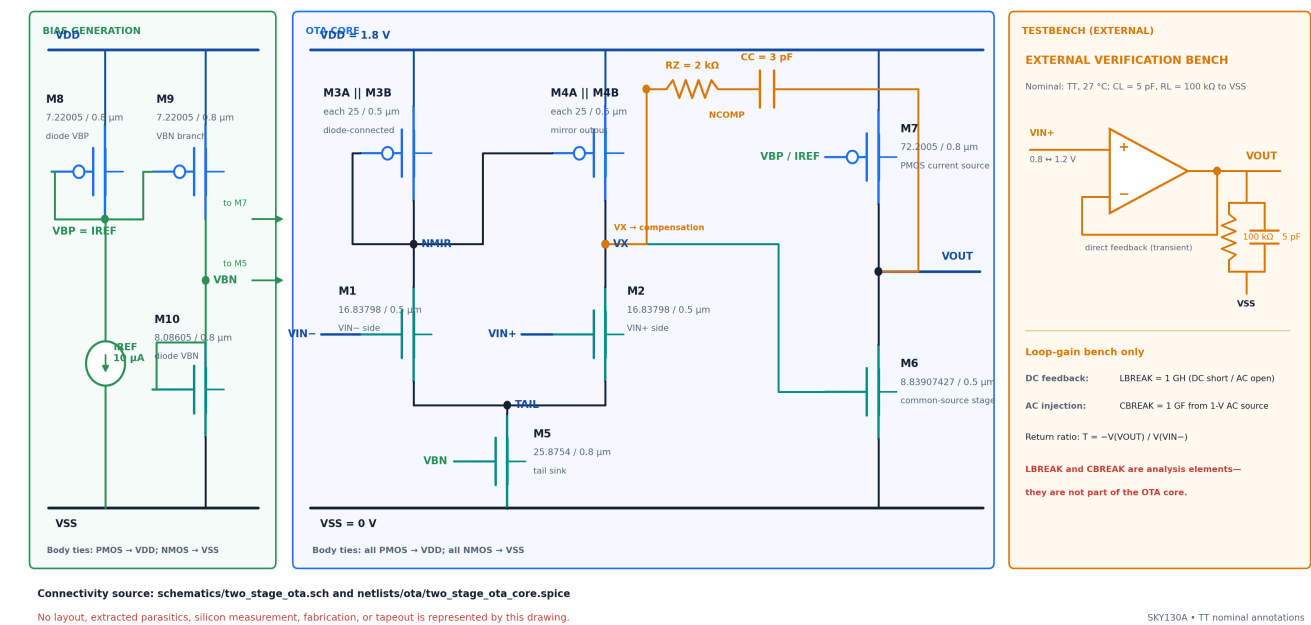


Figure 1. Verified Xschem transistor-level OTA core and pin interface.

Final device geometry

Devices	Function	Geometry (W/L, μ m)	Nominal role
M1 / M2	NMOS input pair	16.83798 / 0.5 each	gm and UGB
M3 / M4	PMOS mirror load	2 x 25 / 0.5 per side	diff-to-single-ended
M5	NMOS tail source	25.8754 / 0.8	38 μ A tail bias
M6 / M7	second stage	8.83907 / 0.5; 72.2005 / 0.8	gain and output drive
M8 / M9 / M10	bias tree	7.22005 / 0.8; 7.22005 / 0.8; 8.08605 / 0.8	10 μ A IREF translation
CC / RZ	Miller network	3 pF / 2 kohm	pole split and zero control

Sizing method

Device lookup tables supplied gm/ID, intrinsic-gain, and VDSAT guidance. Block-level iterations then checked current symmetry, gain, headroom, and area/bandwidth tradeoffs before the complete OTA was tuned. At nominal, all M1-M10 saturation margins are positive; the minimum is 0.116 V at M5.

2. Nominal compensation and transient response

Loop gain uses a DC-closed/AC-open break: a 1 GH inductor preserves follower bias while a 1 GF capacitor injects the AC perturbation. The signed return ratio is $T=-V_{OUT}/V_{INN}$; UGB is the first downward 0 dB crossing and PM is evaluated there.

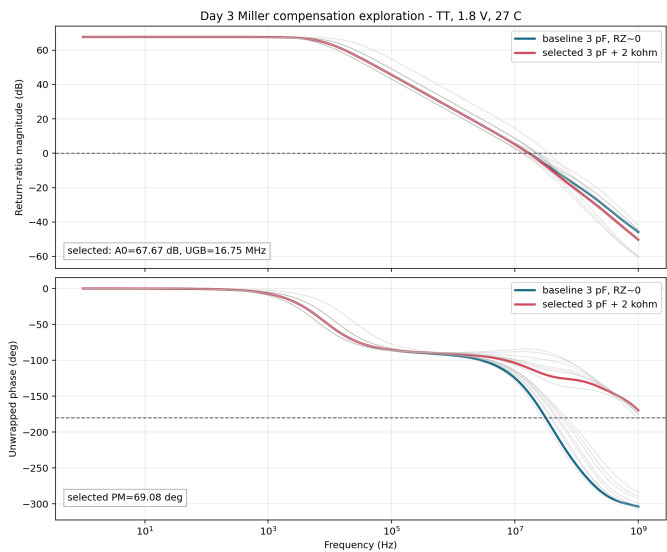


Figure 2. Compensation sweep and selected loop response.

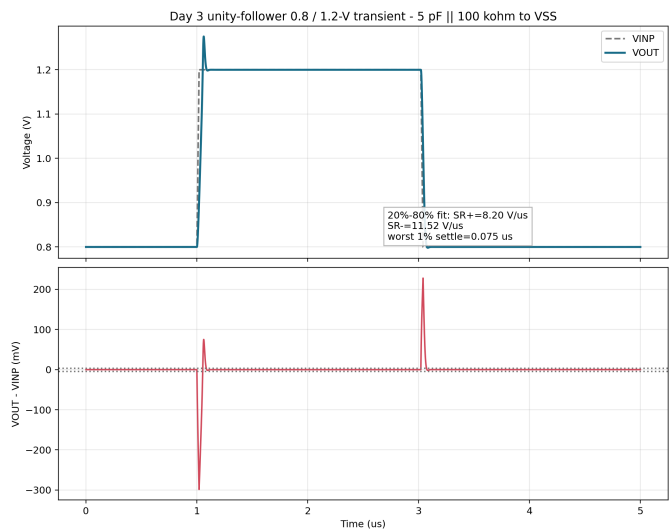


Figure 3. 0.8-1.2-0.8 V unity-follower transient.

Selected nominal result

Metric	Nominal	Hard / stretch	Status
A0	67.68 dB	$\geq 50 / 60$ dB	PASS
UGB	16.75 MHz	$\geq 5 / 10$ MHz	PASS
Phase margin	69.08 deg	$\geq 55 / 65$ deg	PASS
Power	270.54 μ W	$\leq 600 / 400$ μ W	PASS
SR+ / SR-	8.20 / 11.52 V/us	$\geq 2 / 4$ V/us	PASS
Worst 1% settling	0.07475 μ s	$\leq 1.5 / 1.0$ μ s	PASS

Documented optimization: nulling resistor

Configuration	A0	UGB	PM	Disposition
CC=3 pF, RZ=0	67.67 dB	17.26 MHz	33.22 deg	FAIL
CC=3 pF, RZ=1 kohm	67.67 dB	16.36 MHz	52.29 deg	FAIL
CC=3 pF, RZ=2 kohm	67.67 dB	16.75 MHz	69.08 deg	PASS / selected

The selected 2 kohm resistor raises nominal PM by 35.86 degrees relative to the capacitor-only case while changing UGB by about -3%. Slew rate is a least-squares fit over directed monotonic 20-80% segments; settling starts at the interpolated input 50% crossing and requires the output to remain inside a 4 mV band.

3. PVT and load robustness

The fixed matrix contains TT/FF/SS/FS/SF at 1.8 V and 27 C plus the TT combinations of 1.62/1.80/1.98 V and -20/27/85 C, with the duplicated nominal point counted once.



Figure 4. All six core metrics pass at every required PVT point.

Metric	Nominal	Worst PVT	Condition	Hard	Status
A0 (dB)	67.68	65.54	P08	>=50	PASS
UGB (MHz)	16.75	14.55	P08	>=5	PASS
PM (deg)	69.08	66.25	P13	>=55	PASS
Power (uW)	270.54	302.71	P13	<=600	PASS
SR+ (V/us)	8.20	7.98	P06	>=2	PASS
SR- (V/us)	11.52	11.20	P08	>=2	PASS

Nominal load sweep

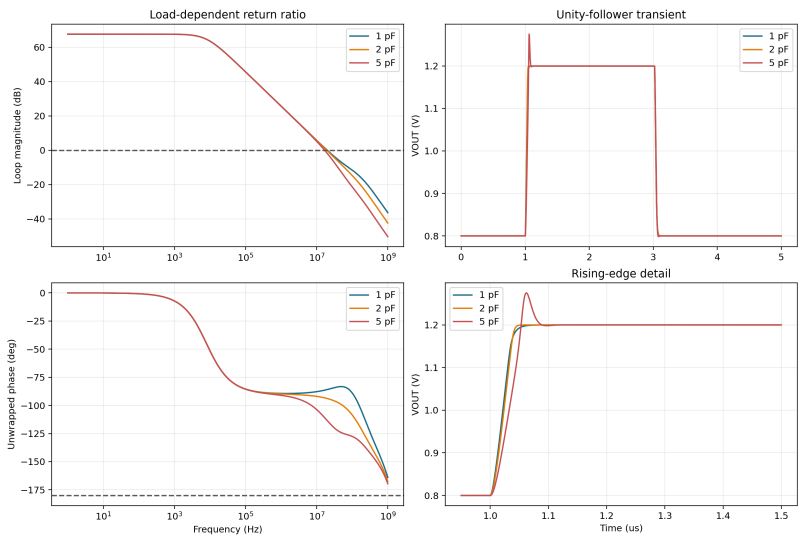


Figure 5. CL=1/2/5 pF with RL=100 kohm. All three PM values exceed 55 degrees and the corresponding unity-follower transients show no sustained or growing oscillation.

Integrity gates

Every required log must finish once without fatal diagnostics; raw grids must have their declared point counts and finite values; loop crossings cannot be ambiguous or land on a sweep boundary. The Day 3 and Day 4 nominal metrics also correlate within their explicit tolerances.

4. Complete nominal characterization

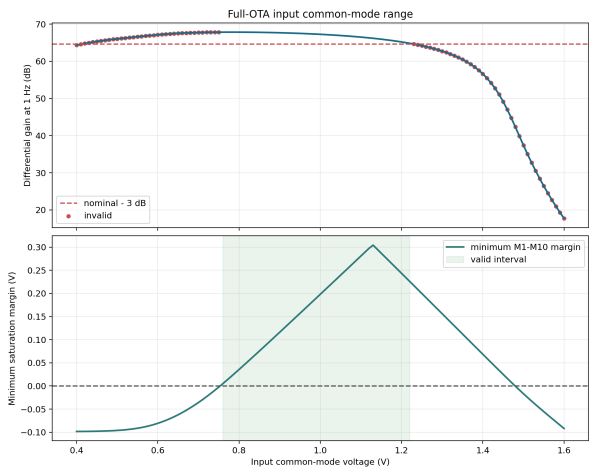


Figure 6. ICMR criteria and limiting interval.

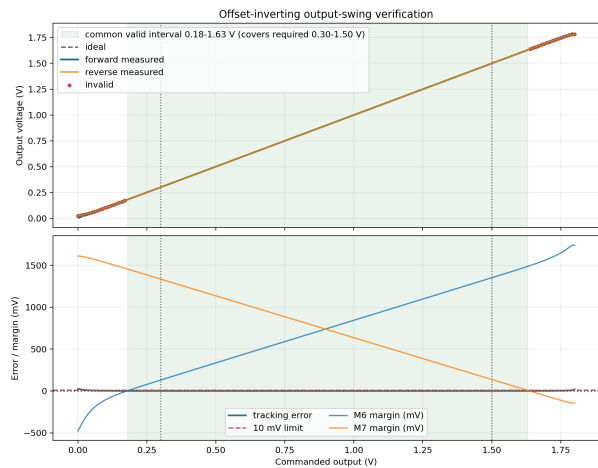


Figure 7. Forward/reverse output-swing audit.

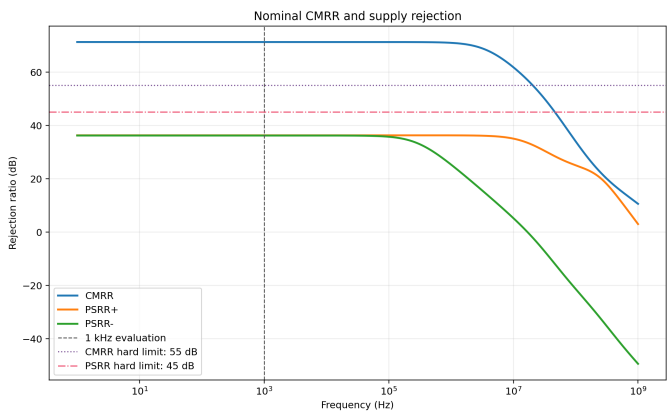


Figure 8. CMRR and both PSRR responses.

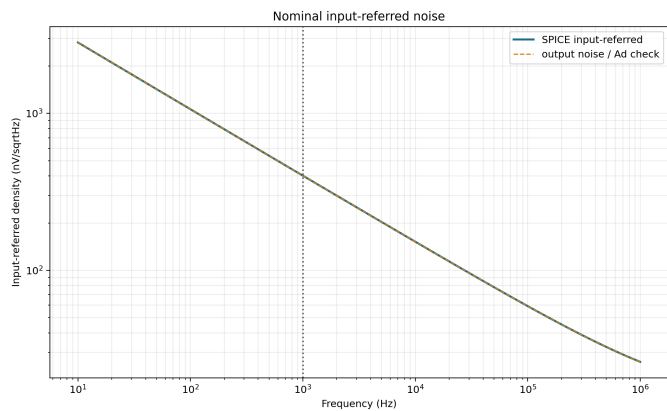


Figure 9. Input-referred noise density.

Metric	Measured result	Requirement	Status
ICMR	0.76-1.22 V	contains 0.8-1.3 V	FAIL
Output swing	0.18-1.63 V	contains 0.3-1.5 V	PASS
CMRR @ 1 kHz	71.32 dB	>=55 dB	PASS
PSRR+ / PSRR- @ 1 kHz	36.33 / 36.25 dB	>=45 dB each	FAIL
Noise @ 1 kHz	401.17 nV/sqrt(Hz)	report	REPORTED
Integrated noise, 10 Hz-1 MHz	52.30 uV RMS	report	REPORTED

ICMR is the largest continuous 10 mV-grid interval around 0.9 V satisfying gain flatness, M1-M10 saturation, and output rail guard criteria. Output swing is measured separately with equal 10 Mohm input/feedback resistors, <=10 mV tracking error, M6/M7 saturation, and agreement between forward and reverse sweeps. Noise is report-only; 24 model conductance-reset warnings are retained in the audit trail.

5. Design decision, limits, and reproduction

Day 5 bounded optimization screen

A nominal-only experiment doubled M1-M4 channel length while preserving W/L. It improved supply rejection, but reduced stability margin, worsened the 1.3 V ICMR checkpoint, and greatly increased device area. It was therefore rejected; the frozen Day 4 geometry remains the reported schematic-level baseline.

Metric	Frozen baseline	M1-M4 Lx2	Change / consequence
A0	67.67 dB	74.38 dB	+6.70 dB
Phase margin	69.08 deg	61.52 deg	-7.56 deg; stretch miss
PSRR+ / PSRR-	36.33 / 36.25 dB	72.42 / 67.32 dB	large nominal improvement
1.3 V ICMR gain delta	-4.94 dB	-7.36 dB	worse by 2.43 dB
Gate-area proxy	1.00x	2.20x	M1-M4 area is 4x
Decision	KEEP	REJECT	REJECTED_TRADEOFF_NOT_PRODUCTION_SELECTED

Reproduce 1. Start Colima/Docker. 2. Use the pinned IIC-OSIC image and SKY130A revision. 3. Run <code>./scripts/run_all.sh</code>. 4. Inspect <code>results/summary.csv</code>, the PVT table, plots, and manifest-stamped raw TSV files. 5. Open <code>schematics/two_stage_ota.sch</code> in Xschem for the editable schematic.	Known limitations - PSRR+ and PSRR- miss the 45 dB hard limit. - ICMR stops at 1.22 V, below the required 1.3 V. - IREF is ideal; reference-generator variation is absent. - No mismatch, Monte Carlo, layout, extraction, package, board, or silicon data. - Non-core metrics were characterized only at nominal.
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Conclusion

This project demonstrates a complete and auditable analog-design workflow: requirements were frozen before tuning, calculations were tied to device data, loop stability was measured with an explicit return-ratio bench, PVT coverage was automated, and unfavorable results were preserved. The design is a credible schematic-level portfolio artifact precisely because its pass boundary and remaining engineering work are explicit.

Primary evidence: README.md | docs/specification.md | results/summary.csv | results/pvt_summary.csv | experiments/day5/assessment.md